Progress in Deep Learning for Remote Sensing Image Classification: A Comprehensive Overview

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Abstract

Deep neural networks have totally changed the way we see satellite and aerial photos. When it works well, new hierarchies can do things that seemed impossible before. They are good at doing hard work about making up scenes, putting things just where they belong, and adding colors to little dots called pixels. The reasons for such success are mainly because of multi-scale feature extraction, paying attention to the important parts, and using information from large collections of data.

But there are also lots of problems. No labeled training samples. Training and inference both take time, so it's hard to fit on a board. Performance degrades when lighting, weather, season, sensors change.

There's lots of fun things for researchers to look at. Lots of people these days are teaching their model using huge bunches of pictures that don't have labels on them instead of putting labels on them by hand. There is a rising trend towards combining optical, radar, elevation and hyperspectral data. On the contrary, there are those who make smaller ones but less accurate. This sort of teamwork makes it possible to expand the real-world applications of this area.

Keywords: deep learning, remote sensing, image classification, convolutional neural networks, scene classification, object detection

1 Introduction

Remote sensing tech has changed a lot in the past ten years. Satellites, planes up there, and even those ground-based sensors take out countless high-resolution pictures every day. There exist different sorts of spectral info, spatial info, and temporal info that can be utilized in land-use mapping, city planning, disaster management, environmental surveillance etc. But as the amount and complexity of remote sensing data has increased, old-fashioned ways of sorting it out have become inadequate.

Many years back, researchers depended largely on handcrafted features. The experts made descriptions by looking at how things seemed through their colors, feels, and shapes. They did alright with simple situations but had trouble when there were messy spots or changing views. Feature Engineering needed lots of info about the subject and much time. It's difficult to get used to different sensors or tasks. Deep learning has flipped all of that on its head. Neural nets learn by looking at raw pixel data without any need for manual feature engineering. CNNs have been the most successful because they can capture spatial hierarchies and context where other models fail.

Remote sensing image is quite different from general image. Generally there are 10 or more spectral channels, and the spatial resolution differs a lot among different platforms. Objects show up on all sorts of scales within this tangled arrangement. Annotation is still costly and time-consuming. And then these special things were able to make it possible for special networks and special ways of training those networks.

This review covers the main events from 2020-2023. Section 2 introduces some basic concepts. Section 3 is about architectural innovation. In section 4, we have 3 major application areas, which are Scene Classification, Object Detection and Semantic Segmentation. In section 5, we discuss current problems and future work, and in section 6, we end the paper.

Table 1: Overview of Deep Learning Applications in Remote Sensing

Application	**Primary Task**	**Challenge**
Scene Classification	Image-level labeling	Multi-scale features
Object Detection	Localization and recognition	Small object detection
Semantic Segmentation	Pixel-level classification	Class imbalance

2 Background

2.1 Fundamentals of Convolutional Neural Networks

CNNs were meant for grids, images, etc. The typical architecture stacks up the convolutional layer, pooling operation, and fully-connected layer, with non-linear activation function interleaved among them.

Convolutional Layer: Feature Detector Slide through the input, find edges, corners, textures, etc. Same filter works everywhere so we don't have many parameters. Pooling layer cuts down on the space of the data but keeps the important parts, this way the model can still figure out what things are no matter where they are and it makes doing calculations less work. Networks form successively more abstract representations as each layer passes along the information. Last but not least, our final fully-connected layer gives us our classification score/bounding box predictions. ReLU has become the default choice of activation functions because it doesn't suffer from vanishing gradient problems and trains faster than other activation functions.

2.2 CNN Training and Optimization

Training process makes it closer to the real label. Backpropagation obtains gradients, then SGD or Adam type optimizers update the weights iteratively. There are plenty of ways to make training more stable and faster.

Data augmentation makes it so that every time it gets rotated, reflected, or colored differently, there's a tiny bit of randomness, which helps the network learn about different things. Dropout and weight decay regularization could prevent overfitting too. Batch Normalization: Activations inside each mini-batch get normalized to make training more stable. Transfer learning works best for remote sensing because we use pre-trained Imagenet weights to start off the network and then do finetuning on smaller datasets.

2.3 Specific Challenges of Remote Sensing

Images usually have over 10 spectral bands, so we need to alter the count of input channels. The ground sampling distance is quite different too which makes the size of objects vary. Scenies are full of meaning, roads, buildings, plants, cars, and water are all combined together. Data sets tend to be a lot smaller compared to general computer vision data set because it takes time and money to label at pixel level or object level. And class imbalance does occur as well;

perhaps there is far more forest than airports.

This implies that the standard ImageNet-trained model doesn't work well so there is effort into domain specific solutions.

3 Progress in CNN Architectures for Remote Sensing

3.1 Multi-scale Feature Extraction

In the objects found in aerial and satellite photographs range from individual cars all the way up to entire cities. Conventional networks with fixed receptive fields tend to lose either the small details or the big picture. There are some good solutions to this problem.

Pyramid Pooling does a pooling operation on lots of various sizes and then puts them back together so it can get both nearby and faraway information. Atrous convolution uses different dilation rates to do similar things but without losing spatial resolution. Feature pyramid networks slowly mix semantically powerful features from lower levels into higher level's high res features using lateral connections creating multiple levels. And the final features are semantically robust at every scale, which is good for finding small things.

3.2 Attention Mechanisms

Now, attention is normal in contemporary architecture. Spatial attention focuses on important places, so it's easier to tell different things apart and ignore the background noise. As for channel attention, from squeeze-and-excitation, we can see that it reassigns feature map according to general importance. Put all these together in modules such as convolution block attention module, this usually gives us the best results.

Take a complicated area such as a port or airport, we can ignore the large uniform parts (water/runways), and focus on the parts that make it different from other areas (ships/aircrafts).

3.3 Transfer Learning and Domain Adaptation

Because there is no public remote sensing dataset that has been annotated, every researcher begins with ImageNet pre-trained backbone. Standard fine-tuning is alright, yet there's still quite some distance between regular pictures and birdseye views. For domain adaptation adversarial approaches and correlation alignment can also help to reduce the gap along with self-supervised learning using pseudo-labels.

3.4 Lightweight Network Architectures

Satellites or drones onboard processing needs networks less than 100MB that infer in several milliseconds. Pruning, Quantization, Knowledge Distillation and Efficient Attention Variants can meet such requirements. Hardware-aware Neural Architecture Search is beginning to create designs for satellite or drone hardware.

4 Challenges and Future Directions

4.1 Limited Training Data

High-quality annotated datasets still is the main problem. Huge sets of unlabelled picture self-supervised pre-training has been given much attention. MoCo, SwAV, DINO and also masked image modelling approaches (MAE, SimMIM) produce reprezentations which can compete, if not outperform ImageNet-supervised learning at least for remotesensing task.

Weaklysupervisedandsemisupervisedapproacheshitokaytoo. For image level labels, rough sketches, or even just points, it doesn't take much to lose out on full pixel masks, but we can still get decent segmentations.

4.2 Computational Efficiency

Real-time onboard inference requires network size¡100MB and inference within tens of milliseconds. Pruning, Quantization, Knowledge distillation, efficient attention mechanisms all do this. Hardware-aware Neural Architecture Search creates networks for satellites and drones.

4.3 Multimodal Data Fusion

Optical sensors don't work at night and when it's cloudy. SAR and LiDAR can work under all kinds of weather but they have little information regarding different colors of lights. How to combine them well, this is still a question. Early fusion is easy but not strong against domain shift. Late fusion is solid but lacks interaction between modes. Intermediate fusion with cross-attention or joint transformers seems to be the way to go.

4.4 Interpretability

Practical application requires stakeholders to know why networks identify an area as ”damaged” following a disaster. Grad-CAM and attention visualization have become standard practice. Concept-based interpretability approaches that map activations to human-understandable concepts such as vegetation indices or building materials are developing but are still immature.

5 Conclusion

2020-2023, great progress. Multi-scale feature fusion + attention mechanism + improved transfer learning + more efficient architecture = practical deployment level. But due to lack of data, computation, multimodal fusion, and interpretability, it still has some problems with its wider application.

Looking ahead, it’s possible that self-supervised and foundation models that have been taught using billions of unlabelled satellite pictures would lessen the necessity for labelling. Better architecture, made especially for edge device. Optical, SAR, Hyperspectral, Temporal data intelligently combined into new applications. Interpretable methods will grow larger, so we can make big decisions with them.

Remote sensing creates a mountain of data that keeps getting bigger every year. Deep learning, modified for this specific region, is still the best way to turn those huge streams of pixels into something useful.

Table 2: Comparison of Deep Learning Methods in Remote Sensing

Method	Advantages	Limitations	Best Applications
CNN-based	High accuracy	High data requirements	Scene classification
Attention-based	Feature selection	High computational cost	Complex scenes
Lightweight models	High efficiency	Potential accuracy loss	Edge deployment
Multimodal fusion	Rich information	Difficult data integration	Fusion tasks

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